

# CONFIGURE INITIAL ROUTER SETTINGS

## 10.1.1 BASIC ROUTER CONFIGURATION STEPS



Complete these steps when configuring initial settings on a Cisco router.

1

Configure the device name.



```
Router(config)# hostname hostname
```

2

Secure privileged EXEC mode.



```
Router(config)# enable secret password
```

3

Secure user EXEC mode.



```
Router(config)# line console 0
Router(config-line)# password password
Router(config-line)# login
```

4

Secure remote Telnet / SSH access.



```
Router(config-line)# line vty 0 4
Router(config-line)# password password
Router(config-line)# login
Router(config-line)# transport input { ssh | telnet }
```

5

Secure all passwords in the config file.



```
Router(config-line)# exit
Router(config)# service password-encryption
```

6

Provide legal notification.



```
Router(config)# banner motd delimiter message delimiter
```

 Use any delimiter (e.g., # or \*) that does not appear in the message.

7

Save the configuration.

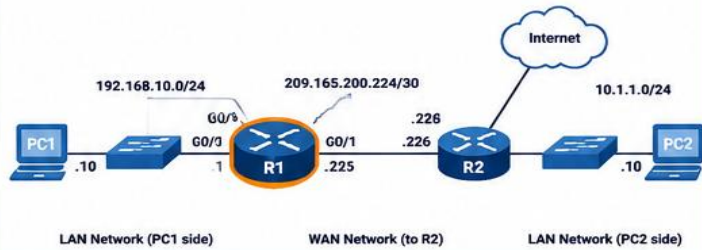


```
Router(config)# end
Router# copy running-config startup-config
```

 Saves the current running configuration so it will be loaded after the router reloads.

## 10.1.2 BASIC ROUTER CONFIGURATION EXAMPLE

### Example Network Topology



### Configure the Device Name

```
Router> enable
Router# configure terminal
Enter configuration commands, one per line.
Router(config)# hostname R1
R1(config)#
```

### Secure Modes (Passwords & Access)

```
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config)#
```

### Encrypt All Passwords

```
R1(config)# service password-encryption
```

### Legal Notification Banner

```
R1(config)# banner motd #
***  UNAUTHORIZED ACCESS IS PROHIBITED ***
This system is for authorized use only.
#
```

 The text between the delimiters (# in this example) is displayed before login.

### Save the Configuration

```
R1(config)# end
R1# copy running-config startup-config
Destination filename [startup-config]? [Enter]
Building configuration...
[OK]
R1#
```



## KEY TAKEAWAYS

1

Name the router



Use hostname for easy identification.

2

Secure access



Use enable secret and passwords for all lines.

3

Restrict remote access



Use transport input ssh to prefer SSH.

4

Encrypt passwords



Use service password-encryption.

5

Show legal banner



Warn users with a clear legal message.

6

Save your work



Use copy running-config startup-config.

7

Secure from threats



Protect against unauthorized access.



**Reminder:** Always secure your router before connecting it to any network. A properly configured router is a secure router!



# CONFIGURE ROUTER INTERFACES

## 10.2.1 CONFIGURE ROUTER INTERFACES



Routers are not reachable until their interfaces are configured and enabled.  
Many interface types exist. Example: Cisco ISR 4321 has two Gigabit Ethernet interfaces.



### Common Interfaces (Example)

- GigabitEthernet 0/0/0 (G0/0/0)
- GigabitEthernet 0/0/1 (G0/0/1)



### Interface Configuration Commands

```
Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown
```



**Note:** When an interface is enabled, a confirmation message should appear.



**Tip:** Use *description* to document the network connected (ISP, service carrier, etc.).  
Max 240 characters.



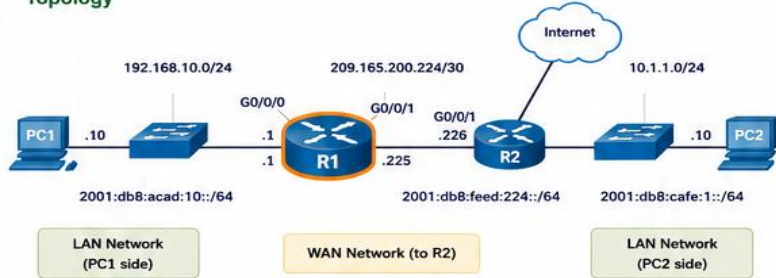
Use *no shutdown* to activate the interface.  
The interface must also be connected to another device (switch/router) for the physical layer to be up.



**Note:** For inter-router links without an Ethernet switch, both connecting interfaces must be configured and enabled.

## 10.2.2 CONFIGURE ROUTER INTERFACES EXAMPLE

### Topology



### Configuration Steps & Commands on R1

```
R1> enable
R1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
%Aug 1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
%Aug 1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
%Aug 1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
R1(config)#
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
%Aug 1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
%Aug 1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
%Aug 1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
R1(config)#
```

**Note:** Information messages show that G0/0/0 and G0/0/1 are enabled and up.

## 10.2.3 VERIFY INTERFACE CONFIGURATION

Useful commands to verify interface settings.

```
R1# show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0/0  192.168.10.1    YES manual up        up
GigabitEthernet0/0/1  209.165.200.225 YES manual up        up
Vlan1          unassigned      YES unset  administratively down down

R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1 [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1            [administratively down/down]
unassigned
R1#
```



### What to Check

- Status = up and Protocol = up for IPv4 interfaces
- IPv6 interfaces show [up/up]
- IP addresses and IPv6 addresses are correct
- VLAN1 is usually administratively down by default

## 10.2.4 CONFIGURATION VERIFICATION COMMANDS

Popular commands used to verify configuration.

| Command(s)                                           | Purpose                                                                                                  |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| show ip interface brief<br>show ipv6 interface brief | Displays all interfaces with IP addresses and their current status. Status and Protocol should be "up".  |
| show ip route<br>show ipv6 route                     | Displays the IP routing table stored in RAM.                                                             |
| show interfaces                                      | Displays statistics for all interfaces (errors, packets, collisions, etc.). Shows only IPv4 information. |
| show ipv6 interfaces                                 | Displays IPv6 statistics for all interfaces.                                                             |
| show ipv6 interface                                  | Displays detailed IPv6 information for a specific interface.                                             |

**Click each command in practice mode to see its output.**

|                                                                                                                                                                                                                                                                                                        |                           |               |                 |                 |                      |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|-----------------|-----------------|----------------------|---------------------|
| show ip interface brief                                                                                                                                                                                                                                                                                | show ipv6 interface brief | show ip route | show ipv6 route | show interfaces | show ipv6 interfaces | show ipv6 interface |
| <pre>R1# show ip interface brief Interface      IP-Address      OK? Method Status  Protocol GigabitEthernet0/0/0  192.168.10.1    YES manual up        up GigabitEthernet0/0/1  209.165.200.225 YES manual up        up Vlan1          unassigned      YES unset  administratively down down R1#</pre> |                           |               |                 |                 |                      |                     |

### KEY TAKEAWAYS



Interfaces must be configured and enabled.  
Routers are unreachable until interfaces are configured and *no shutdown* is issued.



IP addressing on interfaces.  
Assign IPv4 and/or IPv6 addresses with proper subnet mask/prefix length.



Verify your configuration.  
Use *show* commands to confirm interfaces are up and addresses are correct.



Document & secure.  
Use *description* and secure access (passwords, SSH) for a reliable network.



ND replaces ARP in IPv6.  
IPv6 uses ICMPv6 Neighbor Discovery for address resolution and router discovery.



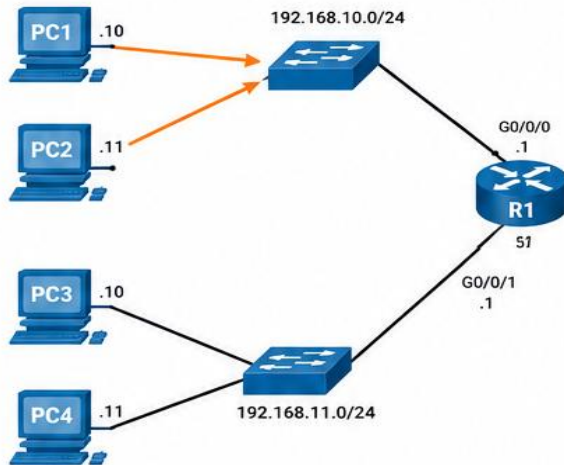
# CONFIGURE DEFAULT GATEWAY

## 10.3.1 DEFAULT GATEWAY ON A HOST

**i** If a host needs to send a packet to a remote network, it sends the packet to its default gateway. The gateway forwards the packet to the next network.

The default gateway on a host is the IP address of the router interface on the local network.

### EXAMPLE TOPOLOGY



### DEFAULT GATEWAY ASSIGNMENT

- Hosts on the same network must use the IP address of the router interface on that network as their default gateway.
- All hosts on a LAN should use the same default gateway.

#### For Hosts in 192.168.10.0/24

**PC3** Default Gateway  
= 192.168.10.1  
(G0/0/0 of R1)

**PC2** Default Gateway  
= 192.168.10.1  
(G0/0/0 of R1)

#### For Hosts in 192.168.11.0/24

**PC3** Default Gateway  
= 192.168.11.1  
(G0/0/1 of R1)

**PC4** Default Gateway  
= 192.168.11.1  
(G0/0/1 of R1)

### KEY POINTS

- ✓ The host forwards packets for remote networks to its default gateway.
- ✓ The default gateway is the IP of the local router interface.
- ✓ Use the correct gateway based on the host's local network.

### BEST PRACTICES



Always set the correct default gateway on hosts based on their local network.



Use the router interface on the same subnet as the default gateway.



Configure an SVI IP on switches only for management access.



Secure remote access (SSH, strong passwords, encryption).



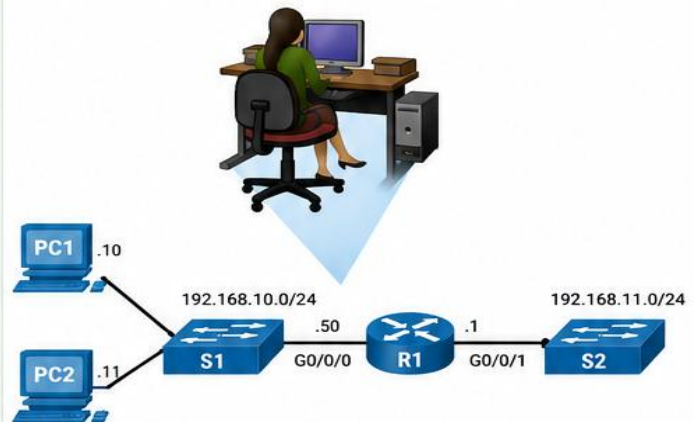
Verify configuration using show commands.

## 10.3.2 DEFAULT GATEWAY ON A SWITCH

**i** A Layer 2 switch does not require an IP address to forward packets. However, an IP address on an SVI (Switch Virtual Interface) is used for remote management.

The default gateway on the switch is the IP address of the router interface on the same local network.

### TOPOLOGY & CONCEPT



### SWITCH CONFIGURATION (EXAMPLE)

```
S1# show running-config
Building configuration....
!
service password-encryption
!
hostname S1
!
interface Vlan1
  ip address 192.168.10.50 255.255.255.0
!
  ip default-gateway 192.168.10.1
!
< Output Omitted >
```

### EXPLANATION

|  |                                        |                                                           |
|--|----------------------------------------|-----------------------------------------------------------|
|  | interface Vlan1                        | Creates/uses the SVI for management.                      |
|  | ip address 192.168.10.50 255.255.255.0 | Sets an IP address for remote management.                 |
|  | ip default-gateway 192.168.10.1        | Sets the router's local interface as the default gateway. |

### HOW IT WORKS

- 1 The administrator host is on a remote network.
- 2 Traffic to the switch's IP (192.168.10.50) is sent to the default gateway (192.168.10.1) on R1 G0/0/0.
- 3 R1 forwards the packet to S1, enabling remote access (SSH/Telnet/HTTPS).